The df_era dataset contains the following columns:

- station_id: A unique identifier for each station, formatted as number1_number2.
- date: The date of the observation.
- tair: Air temperature in Kelvin(although mentioned as Celsius by the client).
- tp: Total precipitation in meters(although mentioned as millimeters by the client).
- sf: Snowfall in meters(although mentioned as millimeters by the client).

There are no missing values

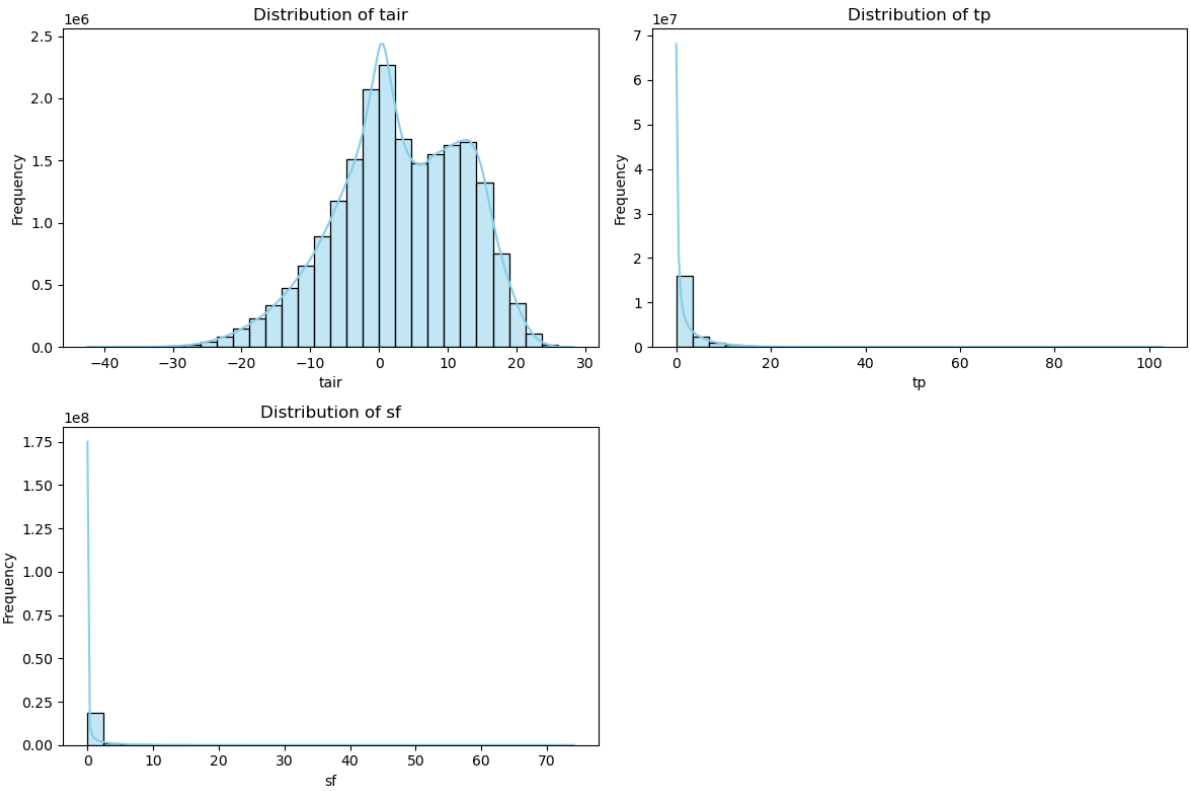
Converted tair to Celsius using the formula:

Celsius=Kelvin−273.15Celsius=Kelvin−273.15

Converted tp and sf to millimeters by multiplying by 1000

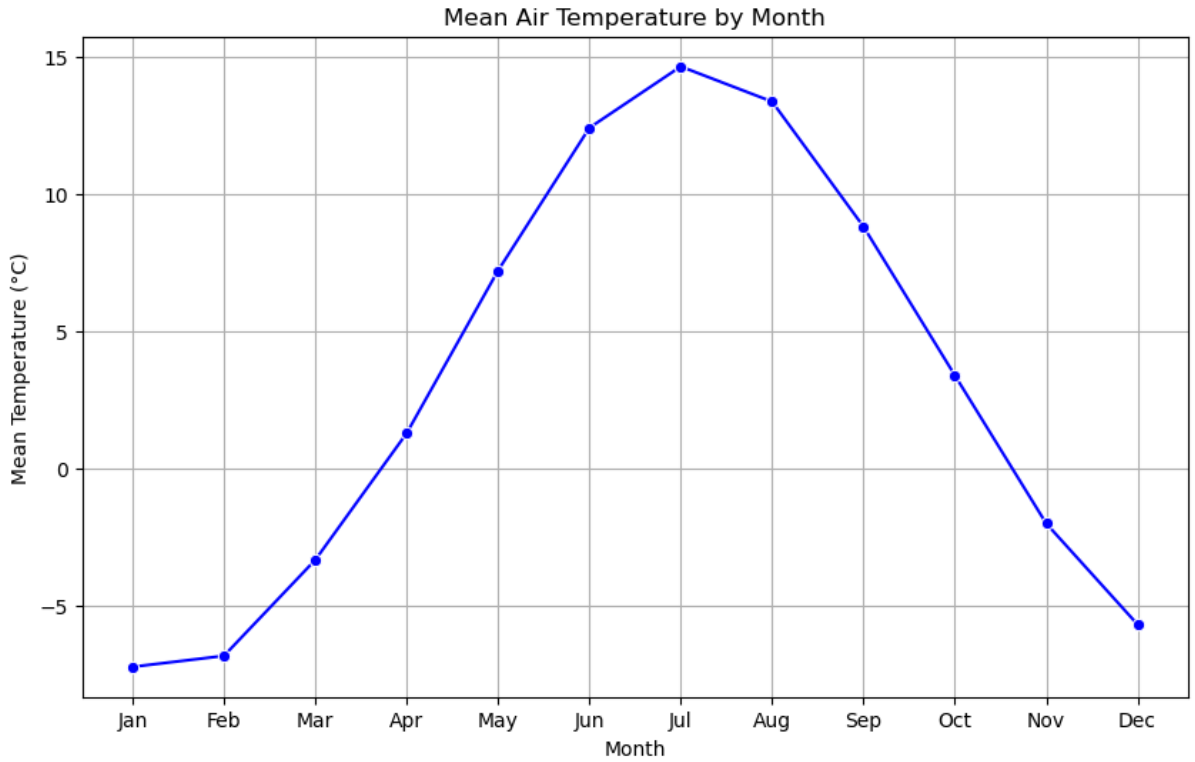
(1 meter = 1000 millimeters).

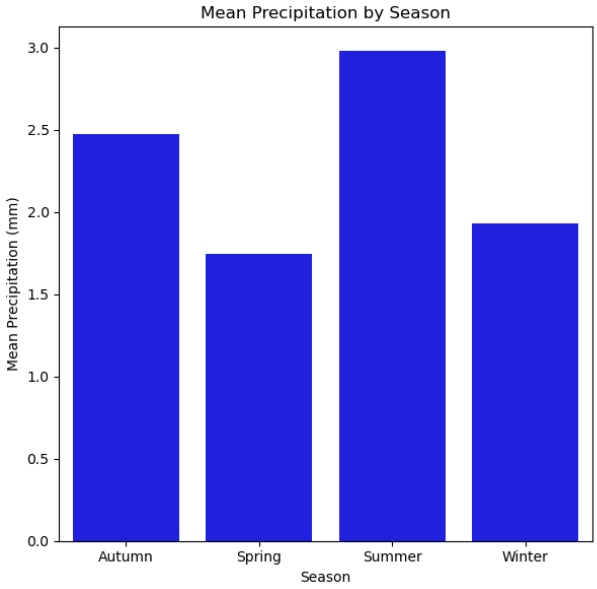
Observations from distribution plots:

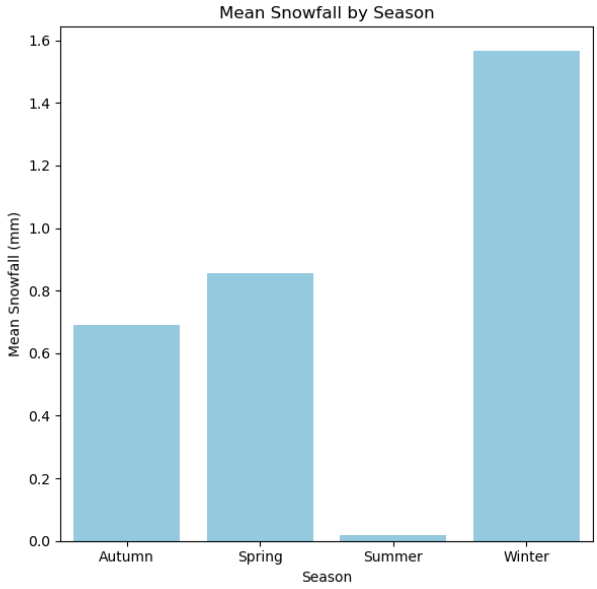


The distribution of tair has two distinct peaks (one at 0°C and another at 13°C) which suggests that the data is bimodal. This is an interesting pattern, and it often indicates the presence of two distinct groups or processes influencing the data.

To explore this further, I analyzed the monthly distribution of *tair*. As expected, January and February were the coldest months, with temperatures rising to a peak in July before declining again.







Summer: High precipitation, almost no snowfall

Winter: Moderate precipitation, peak snowfall

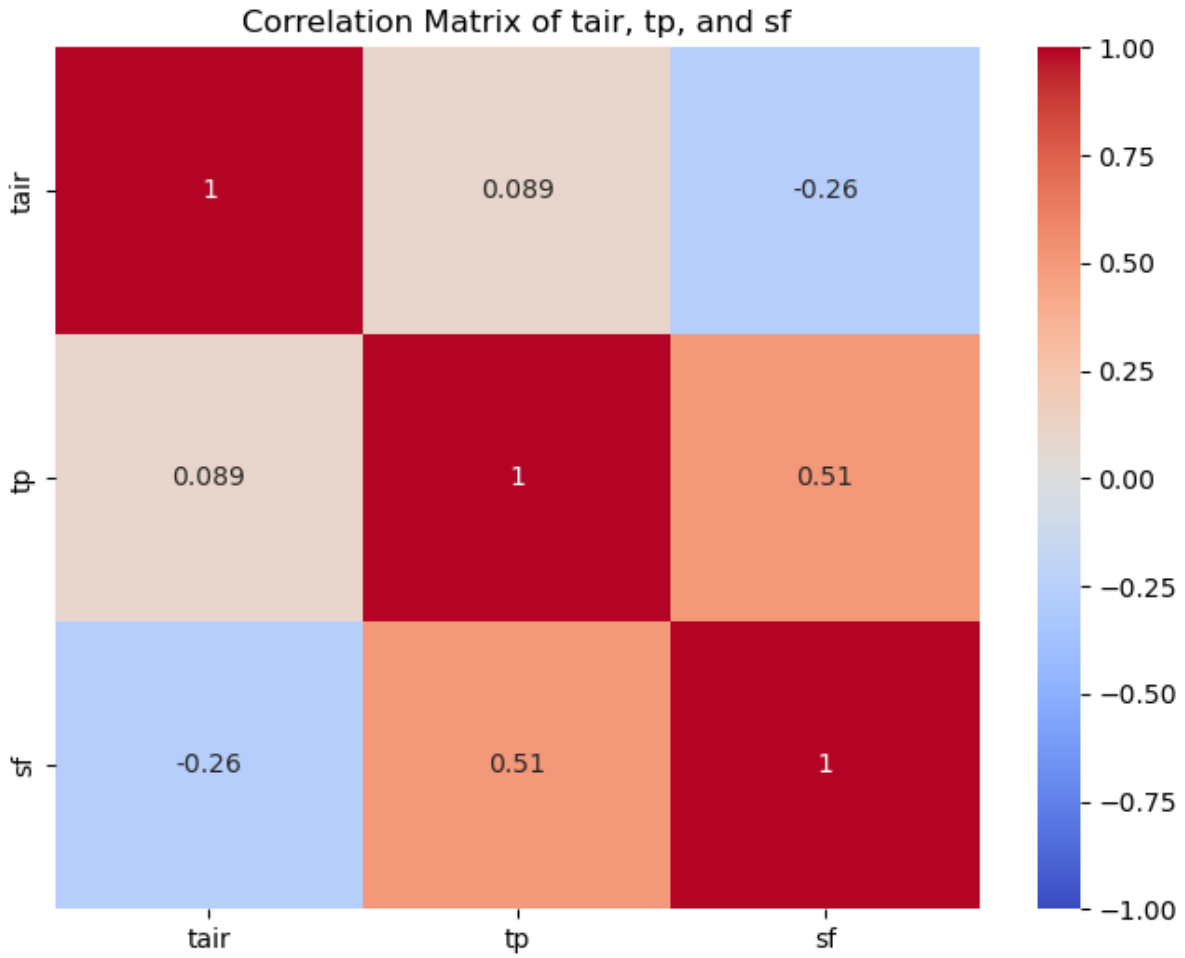
Spring & Autumn: Transition periods

The bimodal temperature distribution likely corresponds to winter(cold peak) and summer(warm peak). The precipitation and snowfall patterns support this:

- Snowfall peaking in winter aligns with the lower temperature mode.
- High summer precipitation but no snowfall aligns with the upper temperature mode.
- Spring and autumn act as transitional seasons, with mixed precipitation types and moderate values, possibly contributing to the less frequent intermediate temperatures.

If the bimodality is strong, it may indicate that temperature rarely stays in a mild mid-range for long, instead oscillating between the extremes of cold and warm. The role of lakes could be relevant here—large bodies of water moderate temperature changes but also provide moisture, possibly amplifying seasonal contrasts.

Correlation analysis:



- tair vs tp – 0.089. This is a very weak positive value, meaning there is almost no linear relationship between temperature and total precipitation. Temperature does not significantly influence total precipitation in the dataset.
- tair vs sf – −0.26. This is a moderate negative correlation, meaning as temperature increases, snowfall tends to decrease. This makes sense, as higher temperatures are less likely to result in snowfall.
- tp vs sf – 0.51. This is also a moderate positive correlation, meaning higher total precipitation is associated with higher snowfall. Snowfall contributes to total precipitation, especially in colder months.

I checked if the correlations between variables change across seasons and this is what I found:

Season	tair vs. tp	tair vs. sf	tp vs. sf
Winter	0.19	0.091	0.89
Spring	0.006	−0.22	0.71
Summer	−0.2	−0.16	0.1
Autumn	0.045	−0.28	0.49

1. Winter:
    a. Snowfall dominates precipitation, with almost all precipitation falling as snow.
    b. Temperature has little influence on snowfall or precipitation.
2. Spring:
    a. Snowfall still contributes significantly to precipitation, but the relationship weakens as temperatures rise.
3. Summer:
    a. Snowfall is rare, and precipitation is mostly rain.
    b. Temperature has little influence on precipitation or snowfall.
4. Autumn:
    a. Snowfall increases as temperatures drop, contributing significantly to precipitation.

Analyzing outliers:

I first checked for outliers across the whole dataset, but seasonal differences resulted in too many. So, I will redo it separately for each season